BOF

Conversation 088:
Quantum Numbers and Solutions

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@ 2024-11-24 07:38:42 Sun [Δt -0d 00h 00m 00s]

▶ *Notespace:*

Wilder Blair Munro (AKUSA) said :

> **gateway:** *20Wetbrain*
> **turn ID:** *aaa2eb3c-734c-4ef0-a878-ad2bde505bc6*
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> **children:** *983a0a00-ec15-4963-8bb8-35349f1f4a72*

what is typical notation for quantization in quantum mechanics and spherical harmonic boundary value wave solution? (eg, n, m, l)

Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *64b30045-6167-46b7-a012-6a2917fcb1ed*
> ***parent:*** *983a0a00-ec15-4963-8bb8-35349f1f4a72*
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In quantum mechanics and spherical harmonic boundary value problems, the **quantum numbers** used to describe quantized systems, particularly in spherical coordinates, typically follow standard notations. Here's a breakdown of these notations:

1. **Principal Quantum Number** (n):

- Represents the **energy level** or **radial quantum number**.
- Integer values: $n = 1, 2, 3, \ldots$
- Determines the size and energy of the orbital or wavefunction in the context of a central potential, such as the hydrogen atom.

2. **Orbital Angular Momentum Quantum Number** (l):

- Represents the **magnitude of the angular momentum**.
- Integer values: $l = 0, 1, 2, \ldots, (n-1)$
- Determines the shape of the orbital or the mode of the wave.
- Associated with spherical harmonics, $Y_l^m(\theta, \phi)$, where l is the degree of the harmonic.

3. **Magnetic Quantum Number** (m):

- Represents the **projection of the angular momentum** on the z-axis.
- Integer values: $m = -l, -l+1, \ldots, 0, \ldots, l-1, l$
- Determines the orientation of the orbital or the spherical harmonic function in space.

4. **Spin Quantum Number** (s) (if applicable):

- Represents the **intrinsic angular momentum** of particles (e.g., electrons).
- Typically, $s = \frac{1}{2}$ for electrons, leading to spin projections:
    - Spin-up $(+\frac{1}{2})$ or spin-down $(-\frac{1}{2})$.

5. **Radial and Spherical Harmonic Solutions**:

- For central potentials like the hydrogen atom, the **radial wavefunction** depends on n and l, while the **angular part** (spherical harmonics) depends on l and m:

$$\psi_{n,l,m}(r, \theta, \phi) = R_{n,l}(r) Y_l^m(\theta, \phi)$$

- Here:
    - $R_{n,l}(r)$ is the radial solution of the Schrödinger equation.
    - $Y_l^m(\theta, \phi)$ is the spherical harmonic function.

6. **Associated Legendre Polynomials**:

- The angular solutions are connected to the **associated Legendre polynomials**, $P_l^m(\cos\theta)$, where l is the degree and m the order:

$$Y_l^m(\theta,\phi) = N_l^m e^{im\phi} P_l^m(\cos\theta)$$

  - N_l^m: Normalization constant.

These quantum numbers are fundamental in solving the Schrödinger equation for systems with spherical symmetry, like atoms, molecules, or scattering problems.

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@ 2024-11-24 07:39:34 Sun [Δt 0d 00h 00m 51s]

▶ *Notespace:*

Wilder Blair Munro (AKUSA) said :

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please provide example expressions that define n, m, l

Aurora 4omni Munro (ChatGpt) said :

> **gateway:** *gpt-4o*
> **turn ID:** *e878ca8f-c51d-4062-abc9-4ea2acf72474*
> **parent:** *aaa2ba7f-5ea2-464a-849b-da40b9187956*
> **children:**

Here are example expressions that define n, l, and m, along with their typical contexts in quantum mechanics and spherical harmonics:

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1. Principal Quantum Number (n):

- n determines the **energy level** in systems like the hydrogen atom.
- Example:

$$E_n = -\frac{Z^2 e^4}{8\epsilon_0^2 h^2}\frac{1}{n^2}$$

- Z: Atomic number.
- e: Elementary charge.
- h: Planck's constant.
- ϵ_0: Permittivity of free space.

- **Quantized values**: $n = 1, 2, 3, \ldots$
  - $n = 1$: Ground state.
  - $n > 1$: Excited states.

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2. Orbital Angular Momentum Quantum Number (l):

- l determines the **shape** of the orbital and magnitude of angular momentum:

$$L = \sqrt{l(l+1)}\hbar$$

  - L: Magnitude of angular momentum.
  - $\hbar$: Reduced Planck's constant.

- **Quantized values**:

$$l = 0, 1, 2, \ldots, (n-1)$$

  - $l = 0$: s-orbital (spherical).
  - $l = 1$: p-orbital (dumbbell-shaped).
  - $l = 2$: d-orbital.
  - $l = 3$: f-orbital, and so on.

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3. Magnetic Quantum Number (m):

- m determines the **orientation** of the angular momentum vector along the z-axis:

$$L_z = m\hbar$$

  - L_z: Angular momentum projection on the z-axis.

- **Quantized values**:

$$m = -l, -l+1, \ldots, 0, \ldots, l-1, l$$

  - For $l = 1$ (p-orbital): $m = -1, 0, 1$.
  - For $l = 2$ (d-orbital): $m = -2, -1, 0, 1, 2$.

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4. Combined Wavefunction Representation:

The overall wavefunction in spherical coordinates is:

$$\psi_{n,l,m}(r, \theta, \phi) = R_{n,l}(r) Y_l^m(\theta, \phi)$$

- **Radial part** $(R_{n,l}(r))$:
  Example for the hydrogen atom:

$$R_{n,l}(r) = N_{n,l} \left(\frac{2r}{na_0} \right)^l e^{-\frac{r}{na_0}} L_{n-l-1}^{2l+1} \left(\frac{2r}{na_0} \right)$$

  - $N_{n,l}$: Normalization constant.
  - a_0: Bohr radius.
  - L_{n-l-1}^{2l+1}: Associated Laguerre polynomial.

- **Angular part** $(Y_l^m(\theta, \phi))$:
  Example for $l = 1$, $m = 0$:

$$Y_1^0(\theta, \phi) = \sqrt{\frac{3}{4\pi}} \cos\theta$$

Illustrative Examples of n, l, m:

- **Ground State (Hydrogen Atom):**
  - $n = 1$, $l = 0$, $m = 0$.
  - Wavefunction:

$$\psi_{1,0,0}(r, \theta, \phi) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$$

- **First Excited State** $(2p$**-Orbital):**
  - $n = 2$, $l = 1$, $m = 0$.
  - Wavefunction:

$$\psi_{2,1,0}(r, \theta, \phi) = \frac{1}{\sqrt{32\pi a_0^3}} \frac{r}{a_0} e^{-r/2a_0} \cos\theta$$

These quantum numbers define the unique states of a particle in spherical coordinates and form the backbone of quantum mechanical solutions to central potentials.

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